Probability

Warm Up

Evaluate.

1. $5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$ 120

2. $7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$ 5040

3. $\dfrac{4 \cdot 3 \cdot 2 \cdot 1}{3 \cdot 2}$ 4

4. $\dfrac{7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{4 \cdot 3 \cdot 2}$ 210

5. $\dfrac{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{(2 \cdot 1)(3 \cdot 2 \cdot 1)}$ 10

6. $\dfrac{8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{(4 \cdot 3 \cdot 2 \cdot 1)(4 \cdot 3 \cdot 2 \cdot 1)}$ 70

To make a yogurt parfait, you choose one flavor of yogurt, one fruit topping, and one nut topping. How many parfait choices are there?

Yogurt Parfait (choose 1 of each)		
Flavor	**Fruit**	**Nuts**
Plain	Peaches	Almonds
Vanilla	Strawberries	Peanuts
	Bananas	Walnuts
	Raspberries	
	Blueberries	

number of flavors	times	number of fruits	times	number of nuts	equals	number of choices
2	×	5	×	3	=	30

There are 30 parfait choices.

Fundamental Counting Principle

If there are n items and m_1 ways to choose a first item, m_2 ways to choose a second item after the first item has been chosen, and so on, then there are $m_1 \cdot m_2 \cdot \ldots \cdot m_n$ ways to choose n items.

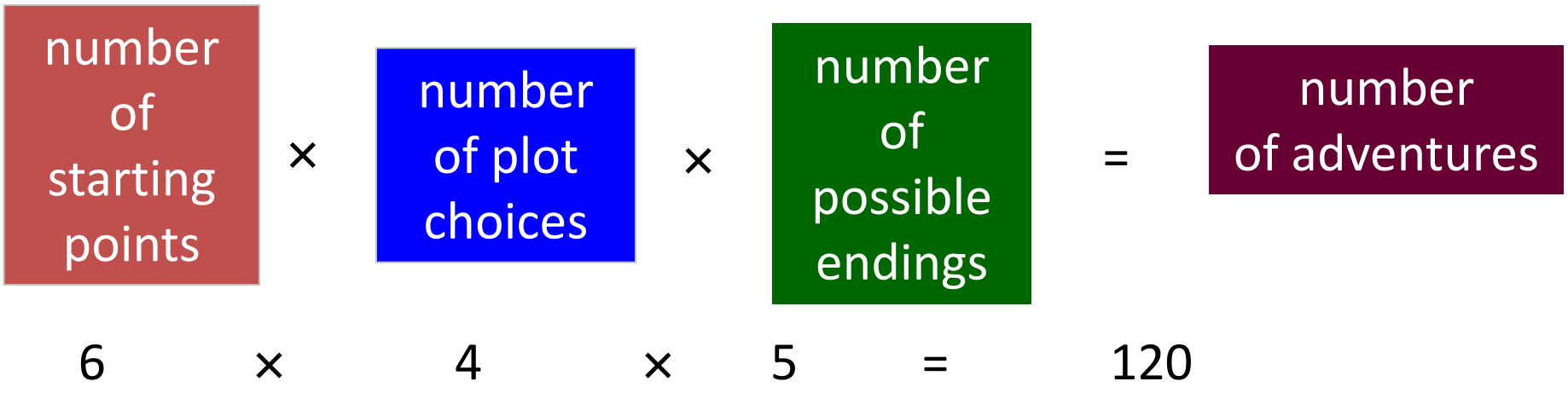
A password for a site consists of 4 digits followed by 2 letters. The letters *A* and *Z* are not used, and each digit or letter many be used more than once. How many unique passwords are possible?

digit digit digit digit letter letter

$10 \times 10 \times 10 \times 10 \times 24 \times 24 = 5{,}760{,}000$

There are 5,760,000 possible passwords.

A "make-your-own-adventure" story lets you choose 6 starting points, gives 4 plot choices, and then has 5 possible endings. How many adventures are there?



There are 120 adventures.

A password is 4 letters followed by 1 digit. Uppercase letters (A) and lowercase letters (a) may be used and are considered different. How many passwords are possible?

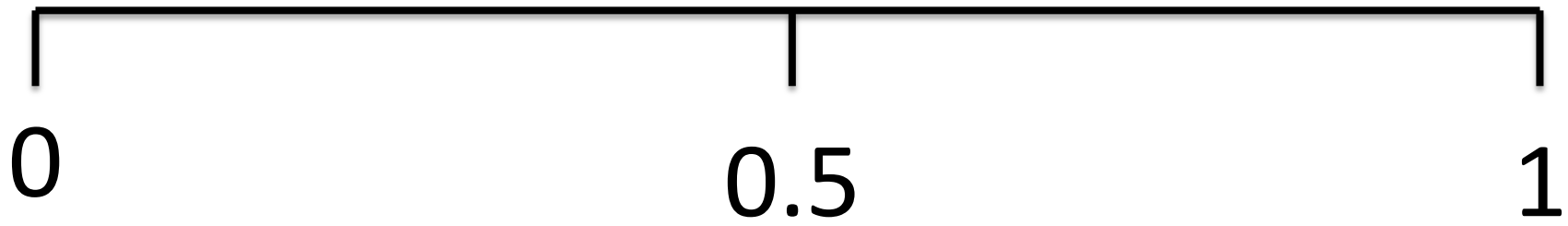
Since both upper and lower case letters can be used, there are 52 possible letter choices.

letter letter letter letter number

$52 \times 52 \times 52 \times 52 \times 10 = 73{,}116{,}160$

There are 73,116,160 possible passwords.

Recap: Probability Scale



Getting a Heads on the flip of a coin

Scoring 101% on a test

Going to sleep tonight

Winning the Lottery

Being left-handed

An IPS student travelling to school by van

How to write probabilities

Probability of winning the lottery:

?

Odds Form

?

Fractional Form

?

Decimal Form

?

Percentage Form

Which is best in this case?

Calculating a probability

$$P(\textbf{event}) = \frac{\text{outcomes matching event}}{\text{total outcomes}}$$

Probability of picking a Jack from a pack of cards?



? = ?

Activity 1 (fill in on your exercise pack)

List out all the possible outcomes given each description, underline or circle the outcomes that match, and hence work out the probability.

The set of all possible outcomes is known as the **sample space**.

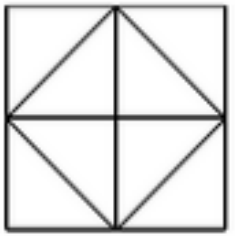
	Event	Outcomes	Probability
1	Getting one heads and one tails on the throw of two coins.	HH, **HT**, **TH**, TT	1/2
2	Getting two tails after two throws.	?	?
3	Getting at least 2 heads after 3 throws.	?	?
4	Getting exactly 2 heads after 3 throws.	?	?
5	Rolling a prime number and throwing a head.	?	?
6	In three throws of a coin, a heads never follows a tails.	?	?
7	For a randomly chosen meal with possible starters **A**vacado, **B**eans and **C**auliflower, and possible main courses **D**eer, **E**scalopes or **F**ish, ending up with neither Avacado nor Dog.	?	?

Puzzle!

Work on in pairs.

[IMC 2009 Q22] A square is divided into eight congruent triangles, as shown. Two of these triangles are selected at random and shaded black. What is the probability that the resulting figure has at least one axis of symmetry?

A $\frac{1}{4}$ B $\frac{4}{7}$ C $\frac{1}{2}$ D $\frac{5}{7}$ E 1



$$= \frac{\text{? Matching}}{\text{? Total outcomes}}$$

Hint: It might be easier to count those which are not symmetrical!

Hint: Can you use the 'choose' function?

Activity 2

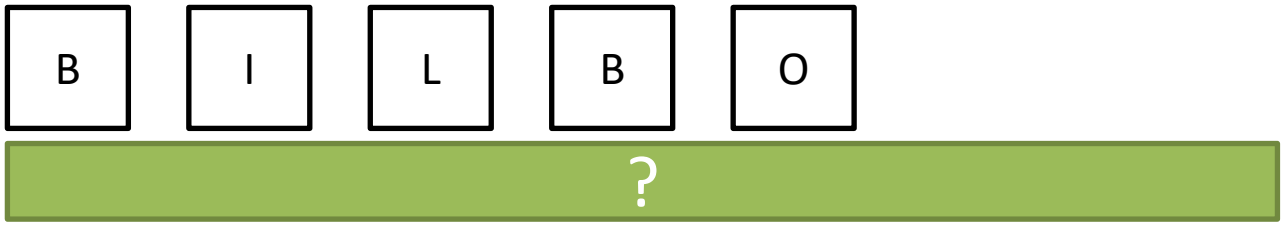
Sometimes we can reason how many outcomes there will be without the need to list them.

	Event	Num matching outcomes	Num total outcomes	Probability
1	Drawing a Jack from a pack of cards.	4	52	P(J) = 4/52 = 1/13
2	Drawing a club from a pack of cards.	?	?	?
3	Drawing a card which is either a club or is an even number.	?	?	?
4	Throwing two sixes on a die in a row.	?	?	?
5	Throwing an even number on a die followed by an odd number.	?	?	?
6	Throwing three square numbers on a die in a row.	?	?	?
7	Seeing exactly two heads in four throws of a coin.	?	?	?
8	Seeing the word ‘BOB’ when arranging two plastic Bs and an O on a sign.	?	?	?
☠	Seeing the word LOLLY when arranging a letter O, Y and three letter Ls on a sign.	?	?	?
☠☠	After shuffling a pack of cards, the cards in each suit are all together.	?	?	?

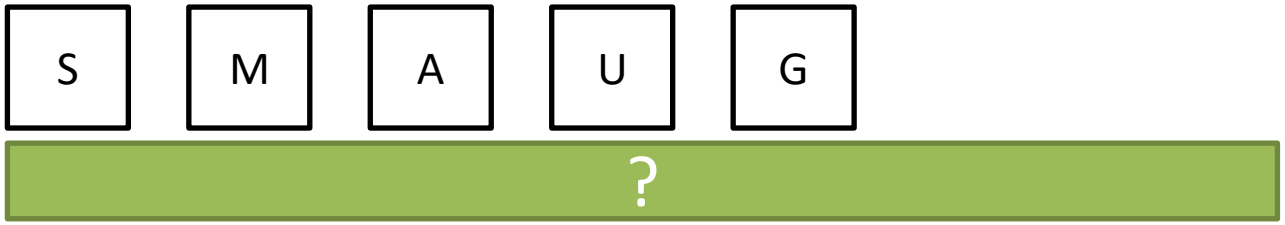
Recap: Combinatorics

Combinatorics is the 'number of ways of arranging something'.
We could consider how many things could do in each 'slot', then multiply these numbers together.

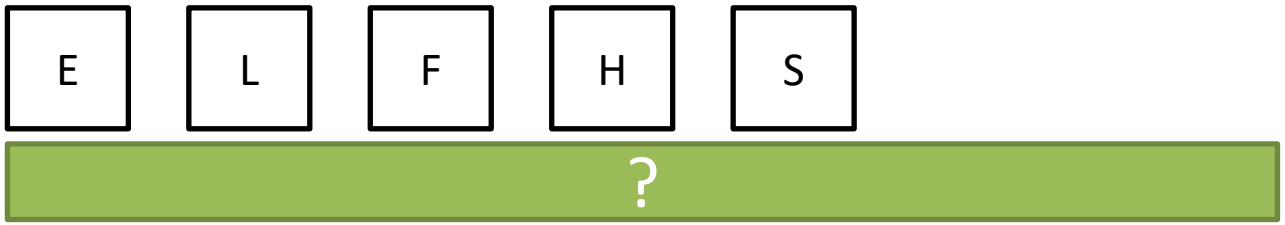
1 How many 5 letter English words could there theoretically be?



2 How many 5 letter English words with distinct letters could there be?



3 How many ways of arranging the letters in **SHELF**?



Activity 3

For this activity, it may be helpful to have four cards, numbered 1 to 4.

	Event	Num matching outcomes	Num total outcomes	Probability
1	One number randomly picked being even.	?	?	?
2	The four numbers, when randomly placed in a line, reads 1-2-3-4	?	?	?
3	Two numbers, when placed in a line, contain a two and a three.	?	?	?
4	Three numbers, when placed in a line, form a descending sequence.	?	?	?
5	Two numbers, when placed in a line, give a sum of 5.	?	?	?
6	When you pick a number out a bag, look at the value then put it back, then pick a number again, both numbers are 1.	?	?	?
☠	When you pick a number from a bag, put the number back, and do this 4 times in total, the values of your numbers form a 'run' of 1 to 4 in any order (e.g. 1234, 4231, ...).	?	?	?

2D Sample Spaces

We previously saw that a sample space was the **set of all possible outcomes**.
Sometimes it's more convenient to present the outcomes in a **table**.

Q: If I throw a fair coin and fair die, what is the probability I see a prime number or a tails?

1D Sample Space

{ H1, **H2**, **H3**, H4, **H5**, H6, **T1**, **T2**, **T3**, **T4**, **T5**, **T6** }

P(prime or T) = 9/12

2D Sample Space

?

2D Sample Spaces

Suppose we roll two ‘fair’ dice, and add up the scores from the two dice.
What’s the probability that:

a) My total is 10?
b) My total is at least 10?
c) My total is at most 9?



Second Dice

First Dice

+	1	2	3	4	5	6
1	?	?	?	?	?	?
2	?					
3						
4						
5						
6						

Three of the outcomes match the event “total is 10”. And there’s 36 outcomes in total.

“At most 9” is like saying “NOT at least 10”. So we can subtract the probability from 1.

Exercise 4

1 After throwing 2 fair coins.



P(HH) = ?

P(H and T) = ?

2 After throwing 2 fair die and adding.



P(total prime) = ?

P(total < 4) = ?

P(total odd) = ?

3 After throwing 2 fair die and multiplying.



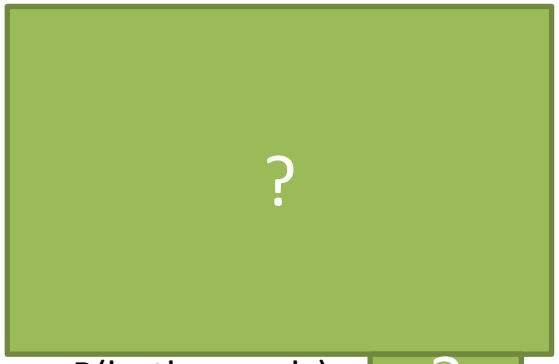
P(product 6) = ?

P(product <= 6) = ?

P(product >= 7) = ?

P(product odd) = ?

4 After spinning two spinners, one A, B, C and one A, B, C, D.



P(both vowels) = ?

P(≥ vowel) = ?

P(B and C) = ?

Events and Mutually Exclusive Events

Examples of events:
Throwing a 6, throwing an odd number, tossing a heads, a randomly chosen person having a height above 1.5m.

An event in probability is ?
(More formally, it is any subset of the sample space)

We often represent an event using a single capital letter, e.g. P(**A**) = 2/3.

If two events A and B are **mutually exclusive**, then ?, and:

P(A **or** B) = ?

You may recall, when we covered Set Theory, that A ∪ B meant "you are in set A, **or** in set B". Since events are just **sets** of outcomes, we can formally write P(A or B) as P(A ∪ B).

Events not happening

A' means that A does not happen.

$P(A') =$?

Quick practice:

1. A and B are mutually exclusive events and P(A) = 0.3, P(B) = 0.2
   P(A or B) = ?
   P(A') = ?
   P(B') = ?

2. C and D are mutually exclusive events and P(C') = 0.6, P(D) = 0.1
   P(C or D) = ?

3. E, F and G are mutually exclusive events and P(E or F) = 0.6 and P(F or G) = 0.7 and P(E or F or G) = 1
   P(F) = ?
   P(E) = ?
   P(G) = ?

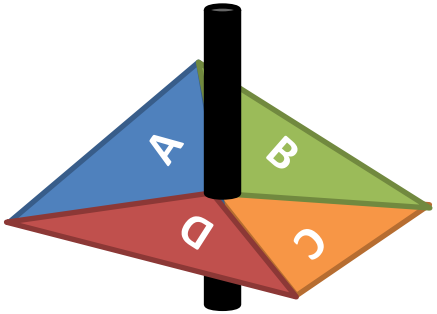
Test your understanding

A bag consists of red, blue and green balls. The probability of picking a red ball is 1/3 and a blue ball 1/4. What is the probability of picking a green ball?

P(G) = ?

An unfair spinner is spun. The probability of getting A, B, C and D is indicated in the table below. Determine x.

A	B	C	D
0.1	x	0.4	x



x = ?

Exercise 5 (on your sheet)

1 In the following questions, all events are mutually exclusive.

a P(A) = 0.6, P(C) = 0.2
P(A') = ? **P(C')** = ?
P(A or C) = ?

b P(A) = 0.1, P(B') = 0.8, P(C') = 0.7
P(A or B or C) = ?

c P(A or B) = 0.3, P(B or C) = 0.9,
P(A or B or C) = 1
P(A) = ?
P(B) = ?
P(C) = ?

d P(A or B or C or D) = 1. P(A or B or C) = 0.6 and P(B or C or D) = 0.6 and P(B or D) = 0.45
P(A) = ? **P(B)**= ?
P(C) = ? **P(D)** = ?

2 All Tiffin students are either good at maths, English or music, but not at more than one subject. The probability that a student is good at maths is 1/5. The probability they are are good at English is 1/3. What is the probability that they are good at music?

P(Music) = ?

3 The probability that Alice passes an exam is 0.3. The probability that Bob passes the same exam s 0.4. The probability that either pass is 0.65. Are the two events mutually exclusive? Give a reason.

?

Exercise 5 (on your sheet)

4 The following tables indicate the probabilities for spinning different sides, A, B, C and D, of an unfair spinner. Work out *x* in each case.

A	B	C	D
0.1	0.3	x	x

x = ?

A	B	C	D
0.5	2x	0.2	x

x = ?

A	B	C	D
x	2x	3x	4x

x = ?

A	B
x	4x + 0.25

x = ?

5 I am going on holiday to one destination this year, either France, Spain or America. I'm 3 times as likely to go to France as I am to Spain but half as likely to go to America than Spain. What is the probability that I don't go to Spain?

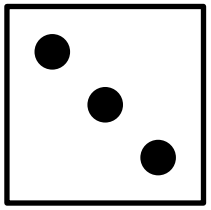
?

☠ P(A or B or C) = 1.
P(A or B) = 4x – 0.1 and P(B or C) = 4x.
Determine expressions for P(A), P(B) and P(C), and hence determine the range of values for x.

?

How can we find the probability of an event?

1. We might just know!



For a **fair** die, we **know** that the probability of each outcome is $\frac{1}{6}$, by definition of it being a fair die.

This is known as a:

Theoretical Probability

?

2. We can do an experiment and count outcomes

We could throw the dice **100 times** for example, and count how many times we see each outcome.

Outcome	1	2	3	4	5	6
Count	27	13	10	30	15	5
R.F.	?					

This is known as an:

Experimental Probability

?

Check your understanding

Question 1: If we flipped a (not necessarily fair) coin 10 times and saw 6 Heads, then $\frac{6}{10}$ is the true probability of getting a Head?

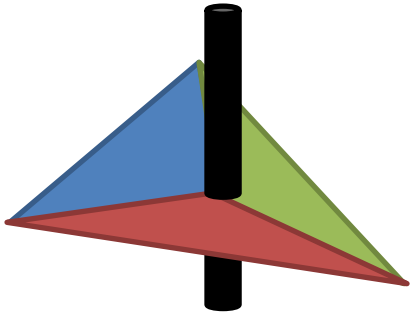
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Question 2: What can we do to make the experimental probability be as close as possible to the true (theoretical) probability of Heads?

?

Estimating counts and probabilities

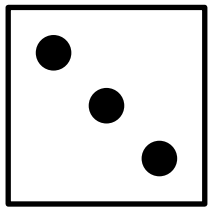
A spinner has the letters A, B and C on it. I spin the spinner 50 times, and see A 12 times. What is the experimental probability for P(A)?



Answer: ?

Answer:

?



The probability of getting a 6 on an unfair die is 0.3. I throw the die 200 times. How many sixes might you expect to get?

Estimating counts and probabilities



The Royal Mint (who makes British coins) claims that the probability of throwing a Heads is 0.4.

Athi throws the coin 200 times and sees 83 Heads. He claims that the manufacturer is wrong.
Do you agree? Why?

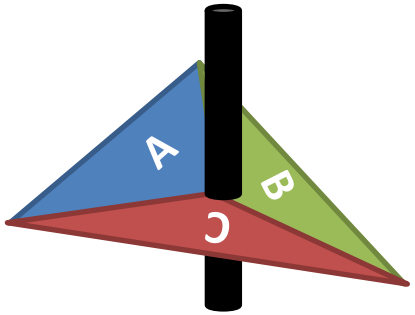
?

Test Your Understanding

A The table below shows the probabilities for spinning an A, B and C on a spinner. If I spin the spinner 150 times, estimate the number of Cs I will see.

Outcome	A	B	C
Probability	0.12	0.34	

?

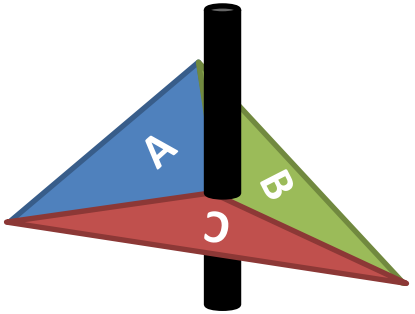


B I spin another spinner 120 times and see the following counts:

Outcome	A	B	C
Count	30	45	45

What is the relative frequency of B?

?



Exercise 6 (on your sheet)

1 An unfair die is rolled 80 times and the following counts are observed.

a) Determine the relative frequency of each outcome.

Outcome	1	2	3	4	5	6
Count	20	10	8	4	10	28
R.F.	?					

b) Dr Bob claims that the theoretical probability of rolling a 3 is 0.095. Is Dr Bob correct?

?

2 An unfair coin has a probability of heads 0.68. I throw the coin 75 times. How many **tails** do I expect to see?

?

3 Dr Laurie throws a fair die 600 times, and sees 90 ones.

a) Calculate the relative frequency of throwing a 1.

?

b) Explain how Laurie can make the relative frequency closer to a sixth.

?

4 The table below shows the probabilities of winning different prizes in the gameshow "I'm a Tiffinian, Get Me Outta Here!". 160 Tiffin students appear on the show. Estimate how many cuddly toys will be won.

Prize	Cockroach Smoothie	Cuddly Toy	Maths Textbook	Skip Next Landmark
Prob	0.37	x	0.18	2x

?

Exercise 6 (on your sheet)

5 A six-sided unfair die is thrown n times, and the relative frequencies of each outcome are 0.12, 0.2, 0.36, 0.08, 0.08 and 0.16 respectively. What is the minimum value of n?

?

6 A spin a spinner with sectors A, B and C 200 times. I see twice as many Bs as As and 40 more Cs than As. Calculate the relative frequency of spinning a C.

?

7 I throw a **fair** coin some number of times and the relative frequency of Heads is 0.45. I throw the coin a few more times and the relative frequency is now equal to the theoretical probability. What is the minimum number of times the coin was thrown?

?

8 I throw an unfair coin *n* times and the relative frequency of Heads is 0.35. I throw the coin 10 more times, all of which are Heads (just by luck), and the relative frequency rises to 0.48. Determine *n*.

[Hint: Make the number of heads after the first throws say , then form some equations]

?